

Leading US geneticist Barbara McClintock's innovative discoveries changed our understanding of genes and their behavior.

Barbara McClintock studied plant genetics, specifically the chromosomes of corn. In 1931, she wrote a paper in which she proposed that chromosomes were the bedrock of genetics, and established herself as the leader in the field. She later discovered that genetic information is not static but can move within or between chromosomes—a phenomenon known as “jumping genes.” Her studies also showed that certain genetic elements can influence the behavior of others. These discoveries were considered too radical until the 1960s, when her results were accepted by her peers. In

1983, McClintock became the first woman to be sole winner of the Nobel Prize in Physiology or Medicine.

“They thought I was **crazy, absolutely mad.**”

Barbara McClintock, 1983

MILESTONES

BECOMES BOTANIST

Graduates from Cornell University with an MSc in 1925, then with a PhD in botany in 1927.

JUMPING GENES

Publishes a paper in 1931 proving chromosomal crossover occurs prior to sexual reproduction.

PLANT GENETICS

Accepts a post in Long Island in 1941; makes pivotal discoveries on inheritance in corn.

LEADER IN FIELD

Becomes the third woman to be elected as a member of the National Academy of Sciences in 1944.

GLOBAL RECOGNITION

Receives the National Medal of Science in 1970, after her groundbreaking work is finally accepted.

McClintock observed that genetic information changes as it is passed down generations, resulting in mutations of physical characteristics.

BARBARA

MCCLINTOCK